



ACID BASE AND SALT

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CHM 100 LEVEL – COMPLETE LECTURE
SLIDE

SLIDE 1: INTRODUCTION

Chemistry deals with substances and their reactions. One of the most important classes of substances in

chemistry are acids, bases, and salts. These substances play major roles in:

- Laboratory reactions
 - Industrial processes
 - Biological systems
 - Everyday life (food, cleaning agents, medicines)
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SLIDE 2: DEFINITION OF AN ACID

1. Arrhenius Definition

An acid is a substance that produces hydrogen ions (H^+) when dissolved in water.

Example:



2. Bronsted–Lowry Definition

An acid is a substance that donates a proton (H^+).

Example:



3. Lewis Definition

An acid is a substance that accepts an electron pair.

Example:



SLIDE 3: TYPES OF ACIDS

(a) Based on Strength

- Strong acids: Completely ionize in water

Examples: HCl , HNO_3 , H_2SO_4

- **Weak acids: Partially ionize in water**

Examples: CH_3COOH , H_2CO_3

(b) Based on Concentration

- **Concentrated acids: High amount of acid per volume**
- **Dilute acids: Low amount of acid per volume**

(c) Based on Basicity

- **Monobasic acid: One replaceable H^+ (HCl)**
- **Dibasic acid: Two replaceable H^+ (H_2SO_4)**
- **Tribasic acid: Three replaceable H^+ (H_3PO_4)**

- Sour taste
- Turn blue litmus paper red
- React with metals to produce hydrogen gas
- React with bases to form salt and water
- Conduct electricity in aqueous solution

Example Reaction:



SLIDE 5: USES OF ACIDS

- H_2SO_4 : Battery acid, fertilizer production
- HCl : Cleaning metals, digestion (stomach acid)

- HNO_3 : Manufacture of explosives and fertilizers
 - CH_3COOH : Vinegar (food preservation)
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SLIDE 6: DEFINITION OF A BASE

1. Arrhenius Definition

A base is a substance that produces hydroxide ions (OH^-) in water.

Example:



2. Brønsted–Lowry Definition

A base is a substance that accepts a proton (H^+).

Example:



3. Lewis Definition

A base is a substance that donates an electron pair.

SLIDE 7: TYPES OF BASES

(a) Based on Strength

- Strong bases: NaOH, KOH
- Weak bases: NH_3 , NH_4OH

(b) Based on Solubility

- Alkalis: Bases soluble in water (NaOH, KOH)
 - Non-alkalis: Bases insoluble in water (CuO, ZnO)
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SLIDE 8: PROPERTIES OF BASES

- Bitter taste

- Slippery or soapy feel
- Turn red litmus paper blue
- Neutralize acids
- Conduct electricity in aqueous solution

Example Reaction:



SLIDE 9: USES OF BASES

- NaOH: Soap making, paper production
- Ca(OH)_2 : Whitewashing walls, water treatment
- NH_3 : Fertilizer production
- Mg(OH)_2 : Antacid (milk of magnesia)

SLIDE 10: DEFINITION OF A SALT

A salt is a compound formed when the hydrogen ion of an acid is replaced by a metal or ammonium ion.

Example:



SLIDE 11: TYPES OF SALTS

- . Normal salts: All H^+ replaced (NaCl)
 - . Acid salts: Partial replacement (NaHSO_4)
 - . Basic salts: Contain OH^- ($\text{Mg}(\text{OH})\text{Cl}$)
 - . Double salts: Two salts crystallized together ($\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$)
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SLIDE 12: PROPERTIES OF SALTS

- Usually crystalline solids
 - Can be neutral, acidic, or basic
 - Many are soluble in water
 - Conduct electricity in molten or aqueous form
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SLIDE 13: PREPARATION OF SALTS

1. Acid + Base $\rightarrow$ Salt + Water
2. Acid + Metal $\rightarrow$ Salt + Hydrogen
3. Acid + Carbonate $\rightarrow$ Salt + CO₂ + Water

Example:



SLIDE 14: pH SCALE

- $\text{pH} < 7 \rightarrow \text{Acidic}$
- $\text{pH} = 7 \rightarrow \text{Neutral}$
- $\text{pH} > 7 \rightarrow \text{Basic}$

Examples:

- $\text{HCl} \rightarrow \text{pH} \approx 1$
- Pure water $\rightarrow \text{pH} = 7$
- $\text{NaOH} \rightarrow \text{pH} \approx 14$

SLIDE 15: CALCULATIVE EXAMPLES

Example 1: Neutralization

Calculate the volume of 0.5 M NaOH needed to neutralize 25 cm³ of 1.0 M HCl.

Solution:

$$M_1V_1 = M_2V_2$$

$$1.0 \times 25 = 0.5 \times V$$

$$V = 50 \text{ cm}^3$$

SLIDE 16: SUMMARY

- Acids donate H^+ ions
 - Bases accept H^+ or produce OH^-
 - Salts are products of neutralization
 - pH scale measures acidity or basicity
 - Acids, bases, and salts are essential in daily life
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SLIDE 17: END OF LECTURE

THANK YOU